UI design

Principle 1: Allow users to maintain control

- **Define interactive modes that allow the user to let them do whatever they want freely**
  - Define different modes and allows easy switching between modes
  - E.g., spell check and editing modes in the word processor, read mode, browser modes, etc.
- **Allow flexible interaction**
  - Allow users to interact via multiple ways like the mouse, keyboard, digitizer pen, voice inputs, multitouch inputs, etc.
- **Allow user interaction to be uninterruptable and undoable**
  - Freely allows the user to quit a work and do another without losing changes
  - Undo/redo also allowed

Principle 1: Allow users to maintain control

- **Allow creating customized operations**
  - Create macros for Frequently occurring operations(e.g., like in MS word)
  - For advanced users to customize UI
- **Hide technical internals from casual users**
  - User is unaware of the underlying OS, file management operations and other complexities
  - E.g., in Application software, there is no need to give OS commands.
- **Design for direct interaction with on-screen objects**
  - Icons or other objects could be dragged or scaled up or down as if they exist physically.

Principle 2: Reduce User's memory load

- **Well designed system doesn't test user's memory**
  - E.g., Use list boxes or checkbox inputs rather than text boxes.
- **Reduce demand to learn more**
  - Users should not have to recall past inputs, rather they are saved and shown using visuals.
- **Establish meaningful defaults**
  - User should have a preference to customize layout or change the default settings.
  - However, reset could change them to defaults again.

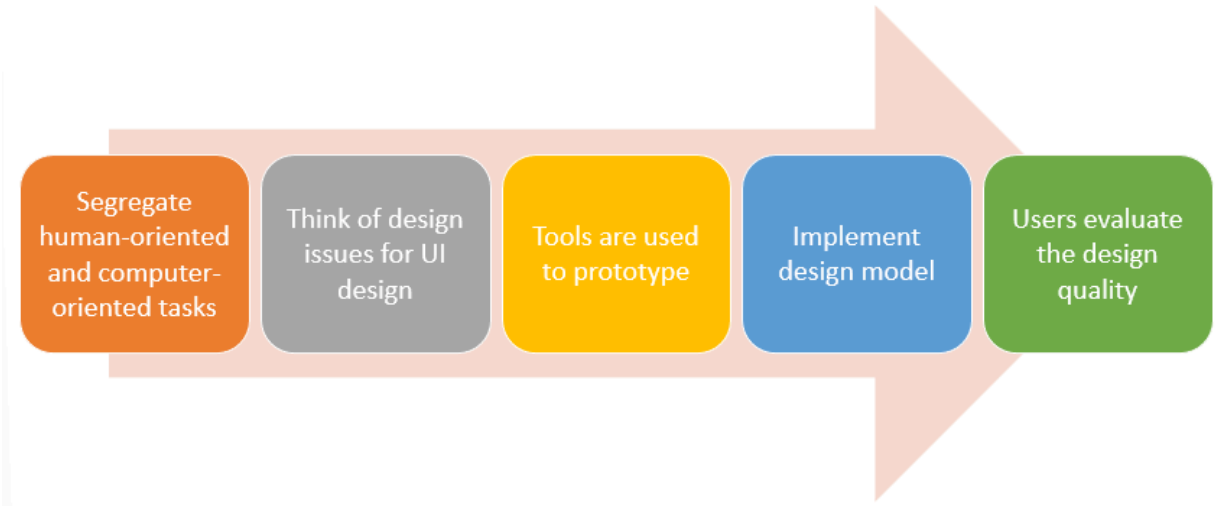
Principle 2: Reduce User's memory load

- **Define intuitive shortcuts**
  - Bind shortcuts that are easy to remember like ctrl+**p** for **print**
- **Visual layout should be near to real-time systems**
  - E.g., the bill paying process should be the same as in real life.
  - All steps must be fulfilled.
- **Disclose information in a progressive fashion**
  - Information displayed on multiple levels.
  - E.g., E-commerce websites have just pictures. Click that to show detailed information, different underline options shown when clicked on the list further

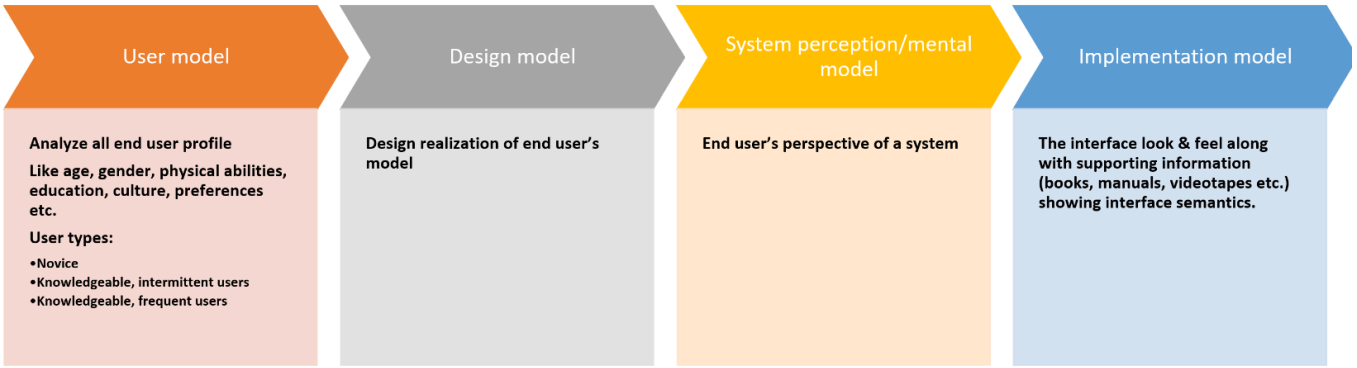
Principle 3: Make consistent interface

- **Allow users to put tasks in meaningful context**
  - User should have an idea of the work they are doing
  - **Like windows should be labeled with titles, consistent color coding should be used for the same tasks etc.**
- **Maintain consistency across a family of applications**
  - A set of applications must follow a particular pattern
  - As the front and the back view is same in whole MS office.
- **Don't make changes in system against user's previous habits, until you are compelled to do so**
  - Don't redefine the user's past experiences in UI
  - For example, Ctrl+S is used for save. Don't bind it with scaling. As user is expecting on his past experience that it'll save the file.

UI Analysis & design



UI Design models



1. User Model

This is the **"Who."** Before a single pixel is drawn, the designer must create a profile of the target audience.

- **User Profiles:** You analyze demographics like age, education, and even physical abilities (e.g., color blindness or motor skills).
- **User Types:**
    - **Novice:** Needs wizards, prompts, and clear labels because they don't know the system.
    - **Knowledgeable, Intermittent:** Knows the basics but forgets specifics; needs good menus and "recognition over recall."
    - **Knowledgeable, Frequent:** Power users who want shortcuts, hotkeys, and fast workflows.

2. Design Model

This is the **"Plan."** It is the designer's vision of how the system should work based on the User Model.

- It incorporates the data structures, the flow of screens, and the "rules" of the interface.
- **The Goal:** To realize the requirements of the user in a way that is technically feasible.

3. System Perception (Mental Model)

This is the **"User's Reality."** It is the image the user forms in their head about how the system works while they are using it.

- If a user clicks a "Save" icon and expects a file to be on their desktop, but it actually goes to the cloud, their **Mental Model** is broken.
- A good UI makes it easy for the user's mental model to match the Designer's Model.

4. Implementation Model

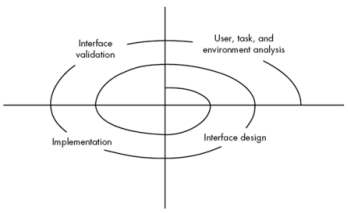
This is the **"Actual Product."** It is the final version that the user touches and sees.

- **Look & Feel:** The colors, fonts, and animations.
- **Semantics:** The "meaning" of the interface—what happens when you swipe, click, or long-press.
- **Supporting Info:** This includes the "Help" section, manuals, or tutorials that help the user understand the system if the UI isn't immediately intuitive.

User Analysis questions for better user categorization

- Are **users trained professionals, technician, clerical, or manufacturing workers**?
- **Level of formal education** of an average user?
- Are the users capable of learning from written materials or have they expressed a desire for classroom training?
- users **expert typists / keyboard phobic**?
- **age range** of the user community?
- User's **gender**?
- How are users compensated for the work they perform?
- Working hours: **normal office hours / until the job is done**?
- **Software usage frequency** ? (mostly/occasionally)
- **primary spoken language among users**?
- **consequences of a user's mistake while using the system**?
- Are users experts in the subject matter that is addressed by the system?
- Are users interested in learning about the technology that sits behind the interface?

UI analysis & design Process



- task analysis:
  - Detailed task analysis done after user analysis.
  - The tasks that users will perform to accomplish system goals.
- User environment analysis:
  - done to check if the current physical environment be able to cater the system needs?
- Output: Analysis model for interface

Interface Design Steps

Interface Analysis

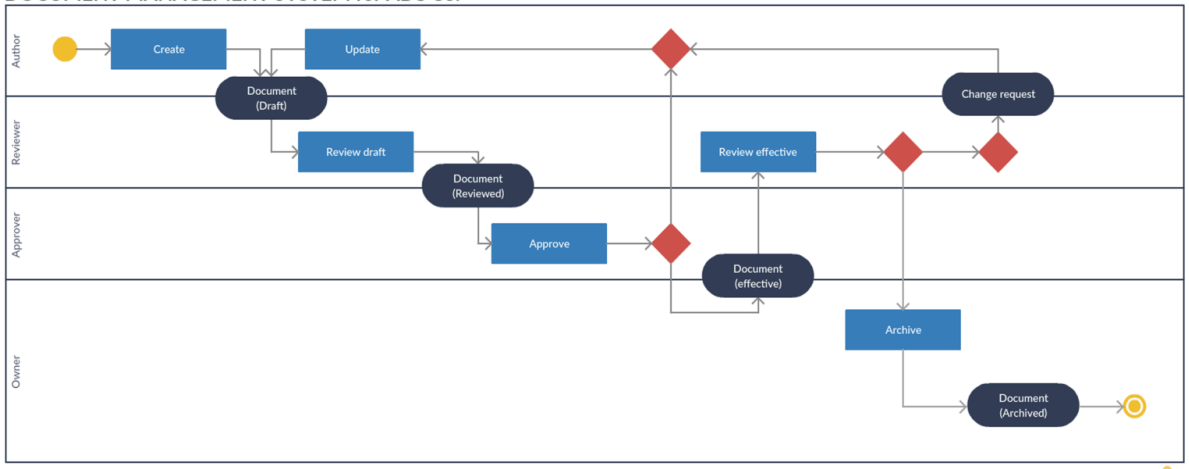
- *Task Analysis & Modelling*
- answer the given questions …
  - Task identification in specific circumstances.
  - Tasks and sub tasks performed by user
  - sequence of work tasks—the workflow
- Use-cases:
  - define basic interaction of actor and system
  - written using informal paragraphs.
  - Can be used to derive tasks, sub tasks and interfaces
- Task elaboration:
  - Stepwise elaboration or task refinement of interactive tasks
  - Derive either manually or use a preexisting system to identify them
- Object elaboration:
  - extract physical objects from system to define their classes and behavior
- Workflow analysis:
  - How defines how a work process is
  - a work process is completed when several people (and roles) are involved
  - Shown using UML swim line diagram

Interface Analysis

- *Task Analysis & Modelling*

Swim line diagram

DOCUMENT MANAGEMENT SYSTEM for ABC Co.



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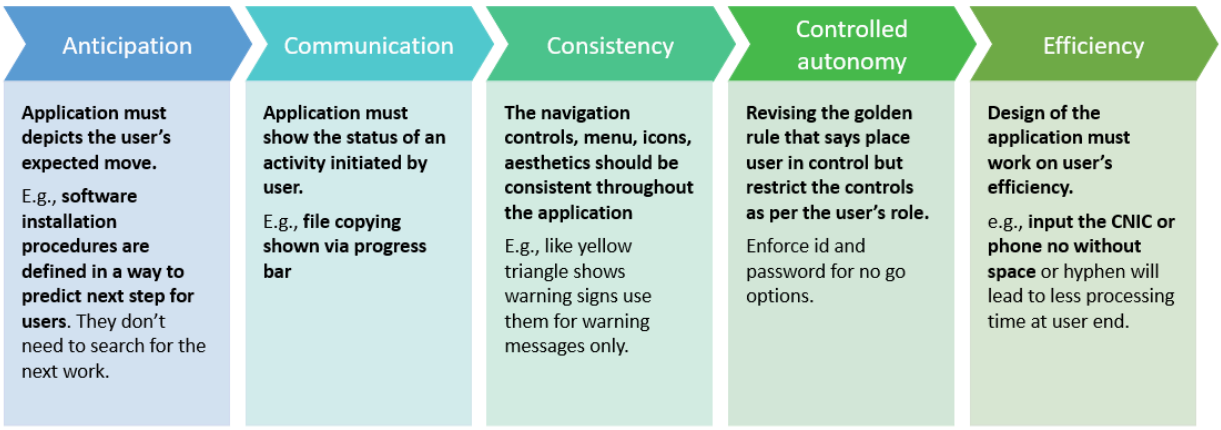
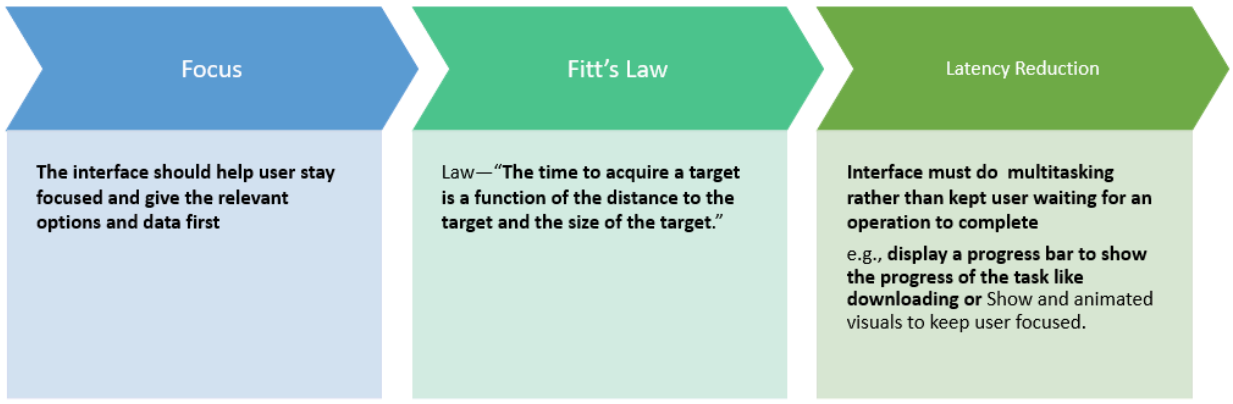
Revised Interface Design Guidelines



Revised Interface Design Guidelines



✅ 1. Anticipation

The system should **predict what the user wants to do next.**

- **Explanation:**

- Design the interface so **users don't have to think** too much.
- **Guide** them **step-by-step.**

🔹 **Example:**

- During software installation:
  - "Next → Accept → Install" flow is obvious.
- User doesn't need to search for the next action.

👉 **Goal: Reduce user effort and confusion.**

✅ 2. Communication

The system should always **inform the user about what is happening.**

🔹 **Explanation:**

- Show **status of ongoing actions**.
- **Provide feedback** for every **user action**.

🔹 **Example:**

- File copying shows a **progress bar**
- "Uploading... 60% complete"

👉 **Goal: Keep users informed and reduce uncertainty**.

✅ 3. Consistency

Everything in the UI should behave and **look uniform.**

Navigation controls,menu,icons,aesthetics shd be consistehnt thru out the application

- **Explanation:**
  - **Same icons, colors, layouts should mean the same thing everywhere.**
- **Example:**
  - ⚠️ **Yellow triangle** always means **warning**
  - Same button style across all pages

👉 **Goal:** Make the system predictable and e**asy to learn**.

✅ 4. Controlled Autonomy

Users should have control, but **within safe limits.**

- **Explanation:**
  - Allow freedom, but **restrict dangerous or unauthorized actions.**
- **Example:**
  - **Require password for sensitive actions**
  - Disable restricted options for certain roles

👉 **Goal:** Balance **freedom + security**

✅ 5. Efficiency

The system should **help users complete tasks quickly.**

🔹 **Explanation:**

- **Minimize unnecessary steps**
- **Reduce typing and effort**

🔹 **Example:**

- Enter CNIC/phone number without spaces or hyphens → faster processing

👉 **Goal:** Save time and effort.

✅ 6. Focus

Keep the user **focused on important tasks.**

🔹 **Explanation:**

- **Show** only **relevant options**
- **Avoid clutter** and distractions

🔹 **Example:**

- Dashboard shows **key data first**
- Hide advanced options unless needed

👉 **Goal: Improve concentration and usability.**

✅ 7. Fitt's Law

The **time to click something {acquire a target}** depends on:

- **Distance to the target**
- **Size of the target**

🔹 **Explanation:**

- **Bigger and closer buttons** are **easier** and **faster to click**

🔹 **Example:**

- Large "Submit" button placed near user's cursor area

👉 **Goal:** Make interaction faster and easier.

✅ 8. Latency Reduction

Reduce waiting time or **make waiting feel shorter.**

🔹 **Explanation:**

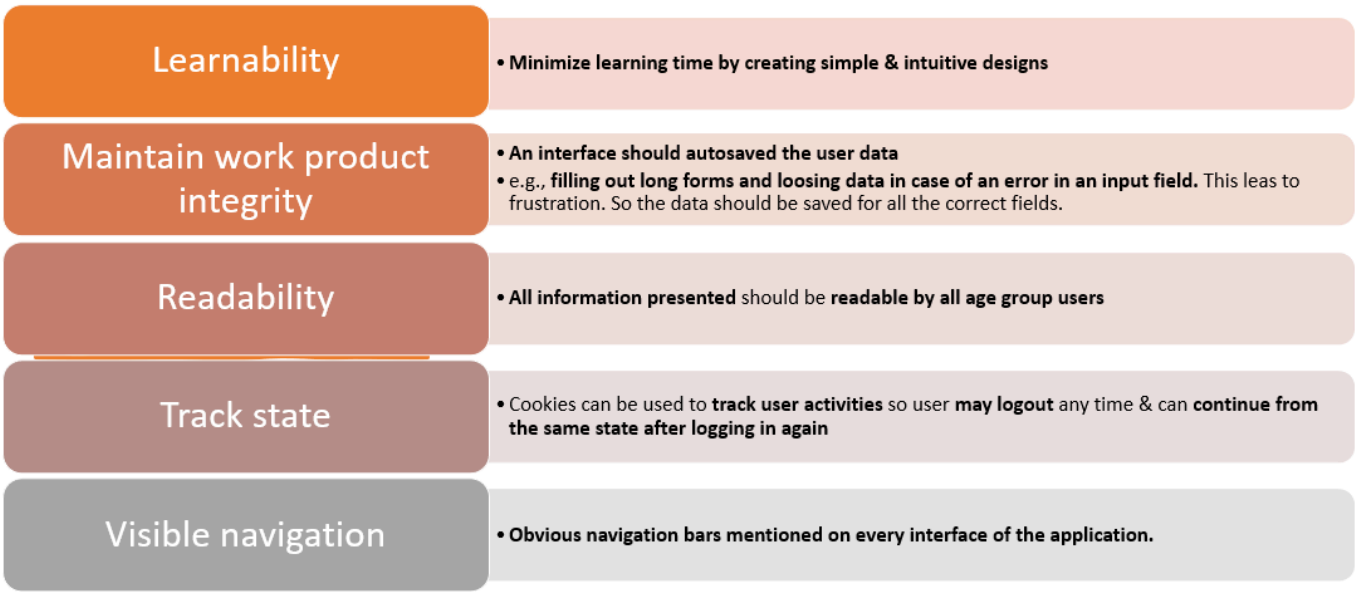
- **Perform tasks in background** (multitasking)
- **Keep user engaged** during delays

🔹 **Example:**

- **Progress bars** during downloads
- **Animations** or **loading indicators**

👉 **Goal:** Improve perceived performance and user experience.

Revised Interface Design Guidelines for a Web App



✅ 1. Learnability

The interface should be **easy to learn fo**r new users.

- Users should understand how to use the system quickly.
- **Avoid complex layouts** or **confusing features**.

🔹 **Example:**

- Simple signup/login process
- Clear buttons like “Next”, “Submit”

👉 **Goal:** **Minimize learning time** and make the system intuitive.

✅ 2. Maintain Work Product Integrity

User data should **not be lost**, even if something goes wrong.

- The system should **automatically save (autosave) user input.**
- Prevent frustration caused by data loss.
- While filling a long form:
    - If an error occurs, already filled correct fields remain saved
- Google Docs auto-saving your work

👉 **Goal: Protect user effort and data.**

✅ 3. Readability

Content should be **easy to read for all users.**

- **Use clear fonts, proper spacing, and simple language**.
- Make it **accessible** for **different age group**s.
- Large readable text
- Good contrast (dark text on light background)

👉 **Goal:** Improve understanding and user comfort.

✅ 4. Track State

The system should **remember what the user was doing.**

- **Track user activit**y so they can **continue later.**
- Often done using **cookies** or **session storage**.
- You log out of a website → log back in → continue from where you left

- Shopping cart items remain saved

👉 **Goal:** Provide **continuity and convenience.**

✅ 5. Visible Navigation

Navigation should be **clear and always accessible.**

- Users should **always know where they are and where to go next**.
- **Navigation menus should be visible on all pages**.
- Top navigation bar (Home, About, Contact)
- Sidebar menus in dashboards

👉 **Goal:** Help users move easily through the application

Interface Design Workflow:

- Derive and refine information from requirements
- Develop rough layout sketch
- Map user objective to actions (like Home, Contact, booking, facililities, etc....)
- Create storyboard screen images for every interface
- Refine layouts by improving aesthetics
- Develop an activity diagram to show the design flow
- Develop state diagrams to show changes in state after transitions
- Describe interface layout for every state
- Refine & review the model.

Interface Design Evolution